

5. Write a program for Implementation of K-Nearest Neighbours (K-NN) in Python

Aim

To implement the **K-Nearest Neighbours (K-NN)** classification algorithm in Python.

Software Required

- Python 3.x
- NumPy
- Scikit-learn

Procedure

1. Load the Iris dataset.
2. Split the dataset into training and testing data.
3. Create a KNN classifier.
4. Select the value of K.
5. Train the classifier.
6. Predict the test samples.
7. Calculate classification accuracy.

Code

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score

iris = load_iris()
X = iris.data
y = iris.target
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = KNeighborsClassifier(n_neighbors=5)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
```

```
print("Predicted Values:")
print(y_pred)
print("\nAccuracy:",
      accuracy_score(y_test, y_pred))
```

Output:

```
... Predicted Values:
    [1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0]

    Accuracy: 1.0
```

Result

Thus, the **K-NN classification algorithm** was successfully implemented and its classification accuracy was obtained.